Engine Model(s): PT6A

73-10-07

PNEUMATIC LINE (P3) - DESCRIPTION AND OPERATION

1. Description and Operation

Compressor discharge air (P3), derived from the diffuser section of the gas generator case, is routed to the metering section of the fuel control unit (FCU) through external lines and, depending on the installation, engine model configuration, through a fine-screen filter.

One configuration utilizes a single heated line, connected directly from a boss on the gas generator case to a metered elbow on the FCU; the line incorporates a flanged elbow at its front end.

A second configuration utilizes two lines with a in-line filter, the body of which is mounted on a bracket at Flange G of the engine; the rear line from the elbow on the FCU to the filter, being of the heated type; the front tube is insulated and has a flanged elbow at its front end.

The third configuration utilizes three lines and a filter, the housing of which is mounted on a bracket at Flange G; the rear heated line from the FCU and the center, unheated line are connected to ports in the filter housing; the front unheated line is interconnected to the center unheated line via a plate mounted on the rear fireseal. Similar to other configurations, the front line has a flanged elbow for connection to a boss on the gas generator case.

All lines, except the flexible ones, are of stainless-steel tubing encased in shrunk-on rubber sheaths. The heated lines incorporates a heater element interposed between the tube and the sheath; a twin-lead cable provides electrical supply from a receptacle mounted on a bracket at Flange G.

The in-line type filter arrangement comprises a ported threaded body and a ported hexagon head, encasing a non-cleanable, life limited 3-micron filter element.

The alternate filter arrangement comprises a manifold-type housing with a removable threaded cover encasing a cleanable, life limited 3-micron or 10-micron filter element. Discharge air enters the housing and is routed via a cored passage to the outer core of the element; filtered air from the inner core of the element is then routed via an internal transfer tube and second cored passage, through a heated tube, to the FCU. Some engine models may have housing with reverse air flow path (Ref. Post-SB1387 / SB1448).

The Post-SB1495 air filter cover has a P3 air pressure sensitive drain valve. During compressor wash, P3 air pressure is low, and the valve is spring loaded open to allow cleaning fluid to drain. As engine speed increases, P3 air pressure increases and closes the valve.

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